

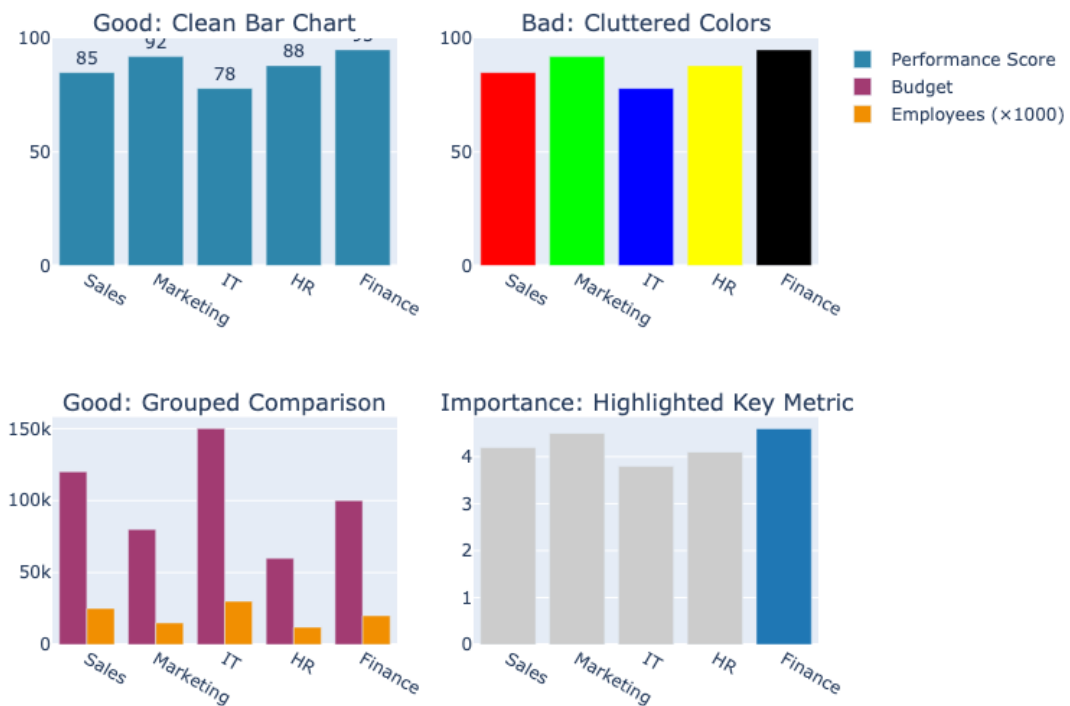
BUS 672 - Data Visualization Types

Comprehensive Examples with Best Practices

This document demonstrates various data visualization types with examples of good practices, common pitfalls, and importance highlighting techniques. Each section covers different data types and their appropriate visualization methods.

Categorical Data Visualizations

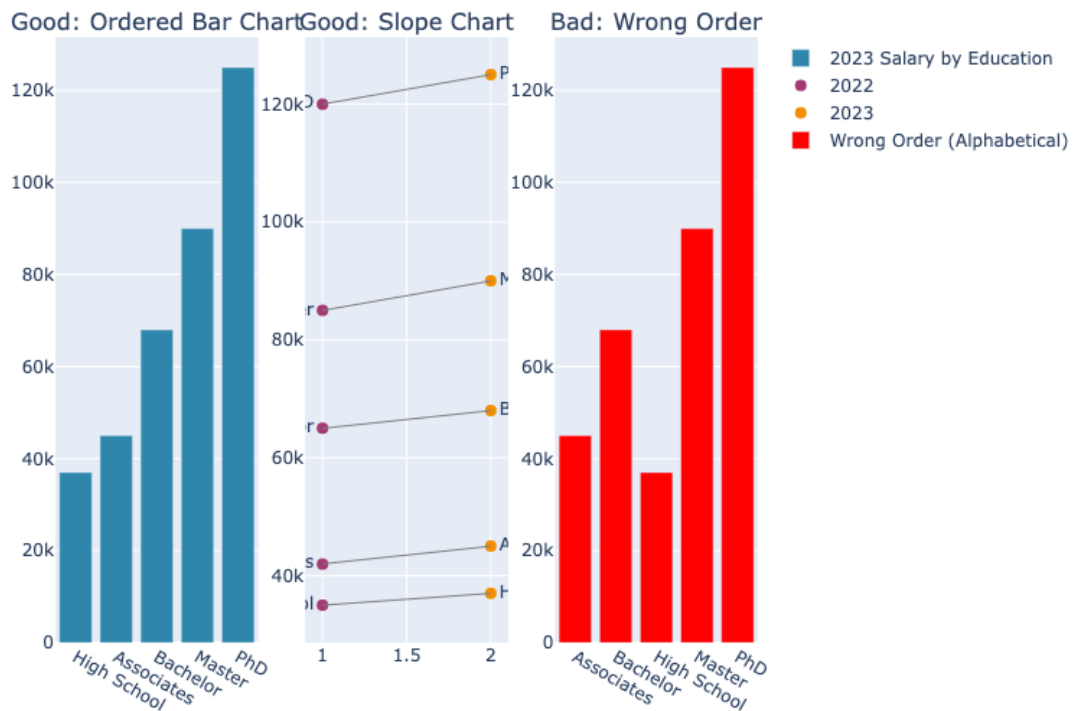
Categorical Data Visualization Examples



- Best Practices: Use consistent colors, clear labels, avoid truncated y-axis
- Common Pitfalls: Too many categories, overuse of colors, misleading scales
- Use Cases: Comparing categories, showing rankings, displaying proportions

Ordinal Data Visualizations

Ordinal Data Visualization Examples

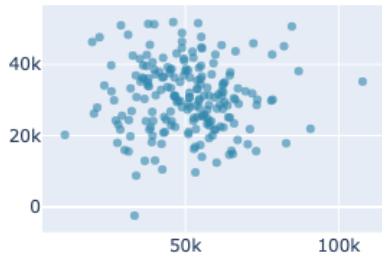


- Best Practices: Maintain logical order, use ordered bar charts or slope charts
- Common Pitfalls: Using alphabetical instead of logical order
- Use Cases: Survey scales, rankings, education levels, time comparisons

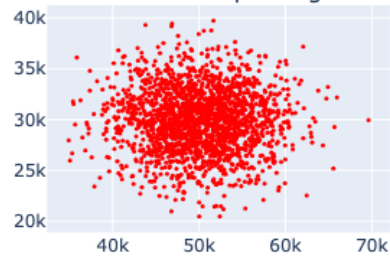
Continuous Data Visualizations

Continuous Data Visualization Examples

Good: Scatter Plot with Correlation

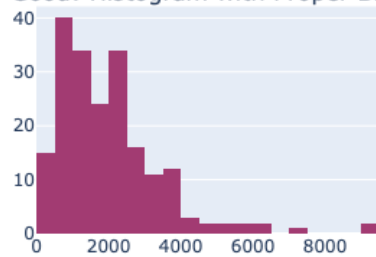


Bad: Overplotting

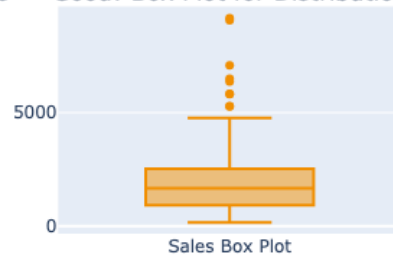


- Income vs Spending
- Overplotted Data
- Sales Distribution
- Sales Box Plot

Good: Histogram with Proper Bins



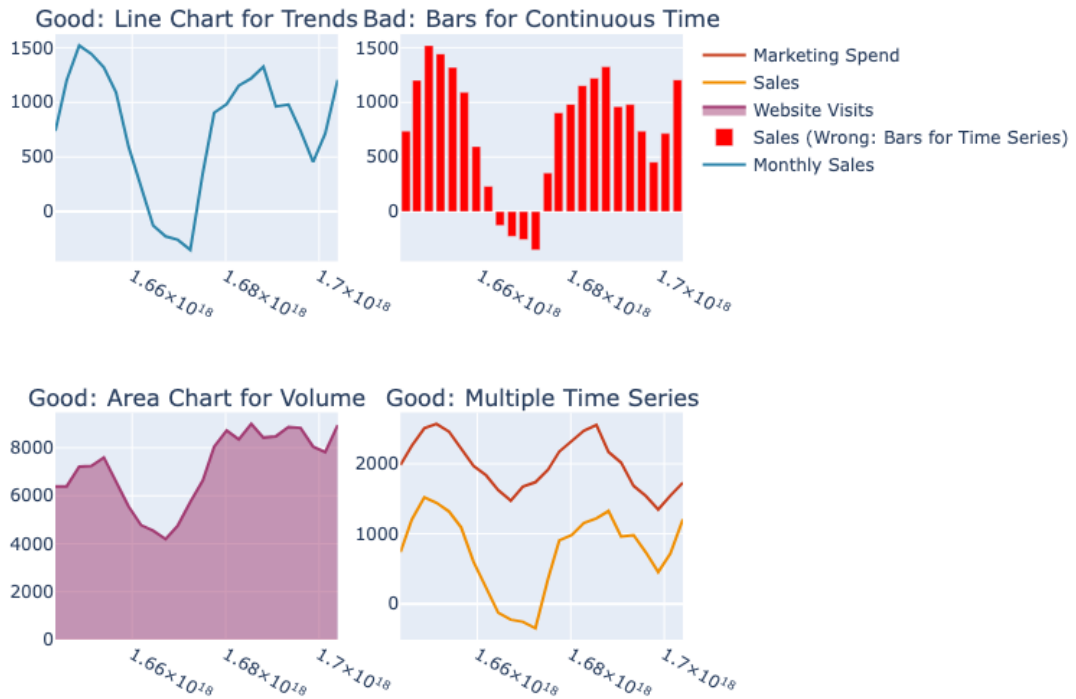
Good: Box Plot for Distribution



- Best Practices: Appropriate bin sizes, clear axis labels, handle overplotting
- Common Pitfalls: Wrong bin sizes, overplotting, unclear scales
- Use Cases: Correlations, distributions, outlier identification

Temporal Data Visualizations

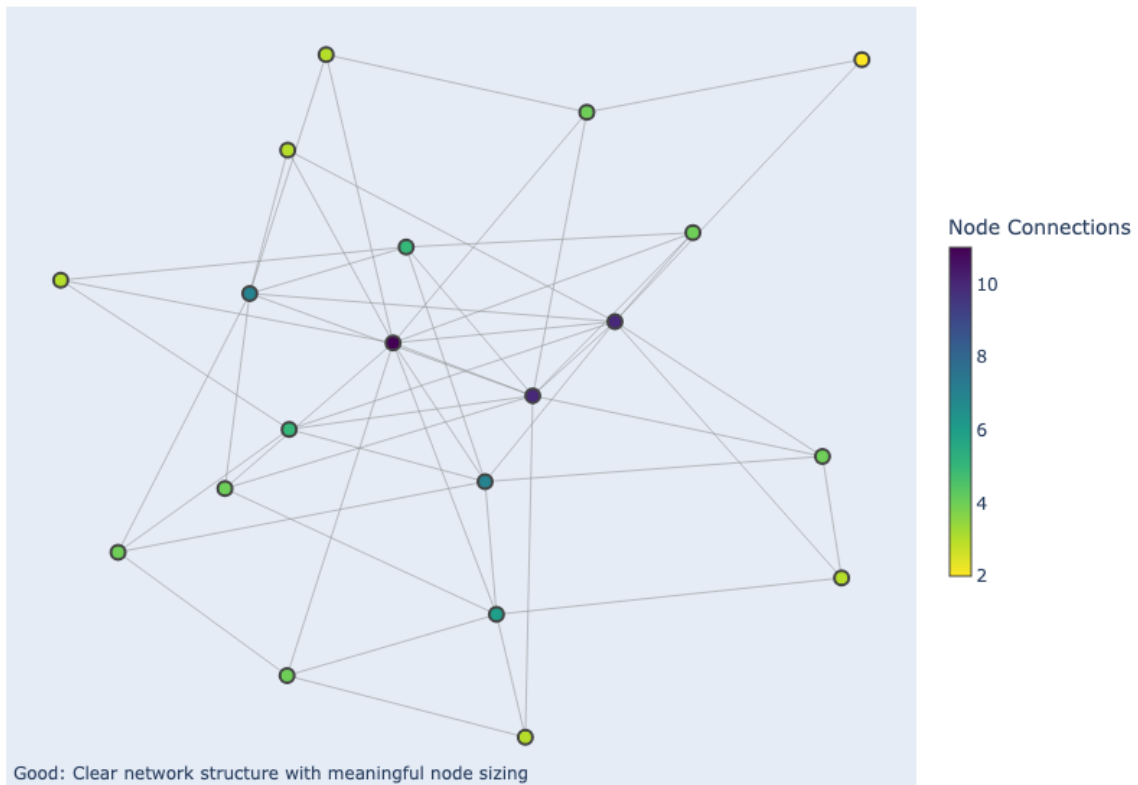
Temporal Data Visualization Examples



- Best Practices: Use lines for continuous time series, show seasonality
- Common Pitfalls: Using bars for continuous time, uneven intervals
- Use Cases: Trend analysis, time comparisons, seasonal patterns

Network/Relational Data Visualizations

Network Diagram Example



- Best Practices: Clear node sizing, meaningful connections, avoid hairball networks
- Common Pitfalls: Overly dense networks, unclear relationships
- Use Cases: Social networks, process flows, entity relationships

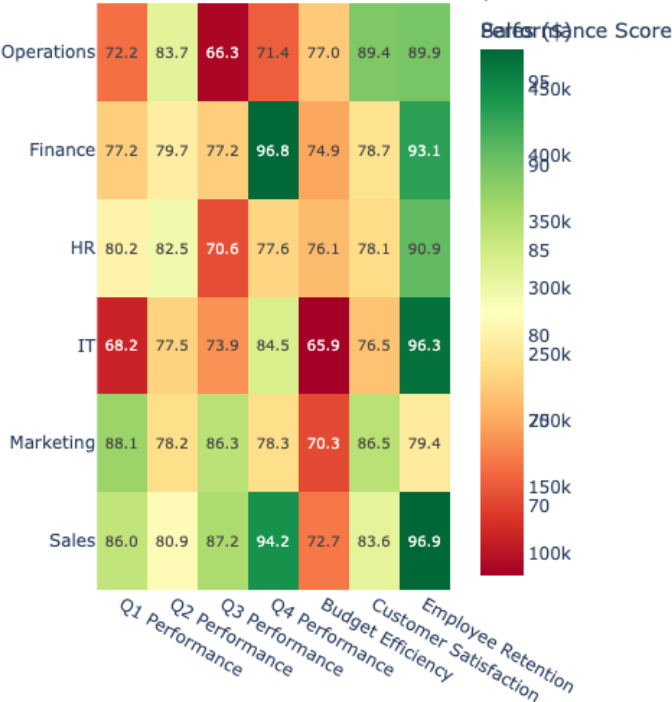
Spatial Data Visualizations

Spatial Data Visualization Examples

Good: Choropleth Map



Good: Business Performance Heatmap



- Best Practices: Appropriate projections, normalized data, clear color scales
- Common Pitfalls: Rainbow color scales, misleading projections, raw vs. normalized
- Use Cases: Geographic comparisons, density mapping, location patterns